# MOSTAFA MAHMOUD ALI

Flutter Developer

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## SUMMARY

Flutter Developer with 3+ years building cross-platform apps across government, logistics and travel — six of them published to Google Play and the App Store, including four consulate apps shipped from a single codebase with 310K+ combined Google Play downloads. Owns feature modules end-to-end — Clean Architecture, BLoC/Cubit, Dio, go_router, get_it — from API contract through flavor setup to store release. Currently the lead mobile engineer on a 23-module consular super-app featuring an SSE-streamed AI assistant, OCR national-ID verification and device-integrity enforcement, backed by 57 automated test suites. Bilingual Arabic/English with full RTL delivery experience.

## TECHNICAL SKILLS

**Languages & Runtime:** Dart 3 (sound null safety), async/await, Futures, Streams, isolates, Kotlin & Swift (platform channels), SQL
**Flutter & UI:** Flutter 3.x, Material 3, Cupertino, custom widgets, responsive & adaptive layouts, animations, RTL/LTR localization (easy_localization, AR/EN), flutter_svg, Lottie
**State & Routing:** BLoC, Cubit, bloc_test, Provider, MVVM, go_router (typed routes, redirects), app_links deep linking, get_it (DI)
**Architecture:** Clean Architecture (feature-first, modular), SOLID, OOP, Repository, Use Cases, Freezed, dartz (Either) error handling
**Data & Networking:** Dio (interceptors, CancelToken, retry), REST APIs, Server-Sent Events (SSE) streaming, JSON serialization, OAuth 2.0 / JWT, pretty_dio_logger, Talker
**Persistence & Storage:** SQLite, flutter_secure_storage, Envied (encrypted build-time secrets), path_provider, offline-first sync
**Native & Device:** Platform channels, Firebase (Auth, FCM, Firestore, Analytics, App Check), Google & Apple Sign-In, local_auth biometrics, root/jailbreak detection, mobile_scanner QR, image_picker / file_picker, share_plus, permission_handler, Google Maps, flutter_map, Stripe & PayPal
**Media & Video:** FFmpegKit (filter graphs, stream mapping, transcoding, speed re-encoding), camera capture (CameraX), video_player, just_audio, HLS / .m3u8 handling, image compression, thumbnail generation
**Testing & Quality:** flutter_test, widget testing, bloc_test, Mocktail, DI test doubles, flutter_lints, Flutter DevTools profiling
**Release & CI/CD:** Multi-flavor builds (dev/staging/prod) with flutter_flavorizr, Play Console, App Store Connect, Play App Signing, build_runner code generation, Git / GitHub, GitHub Actions (CI/CD release gates)

## PROFESSIONAL EXPERIENCE

**Flutter Developer — AppRock Software Development** Aug 2024 – Present
*Remote, full-time*

- Own feature modules **end-to-end** across a 23-module Flutter codebase — entities, use cases, repositories, Cubits and UI — under a feature-first **Clean Architecture** with **get_it** dependency injection.
- Built a real-time AI assistant on a custom **Server-Sent Events** client over **Dio**, with CancelToken-backed stream cancellation, a 3-minute idle watchdog and last-event-id resumption so dropped sockets never leak or duplicate a conversation turn.
- Shipped **OCR-driven national-ID verification** that reconciles front and back card extractions against user-typed input and blocks mismatched submissions client-side before they reach back-office review.
- Hardened the app against tampering with **root/jailbreak device gating**, **biometric authentication** (local_auth), **Firebase App Check**, Envied-encrypted build secrets and QR-based device linking.
- Grew automated coverage to **57 test suites** using flutter_test, bloc_test and Mocktail across Cubits, repositories, SSE frame parsing, deep-link routing and widget behaviour — catching 3 defects before release.
- Delivered **3 build flavors** (dev/staging/prod) with separate application IDs, Firebase projects and signing configs; diagnosed a **Play App Signing / Firebase SHA-1 fingerprint mismatch** that was blocking Google Sign-In and App Check on the internal track.
- Handle end-to-end release into **Google Play Console** and **App Store Connect** — keystore and signing setup, version and build numbering, store listings and screenshots, internal and closed test tracks, TestFlight distribution, staged rollout and app review submissions across both platforms, promoted via GitHub Actions release gates.
- Implemented deep linking, FCM push with actionable notifications, QR generation and scanning, and an image/PDF upload pipeline with client-side compression; partner directly with backend and design on API contracts and Material 3 handoffs.

**Flutter Developer (Freelance)** Mar 2023 – Present

*Remote, part-time*

- Delivered client Flutter apps end-to-end — translating designs into production UI, integrating REST APIs and payment providers, and publishing to both stores under **BLoC**, **Freezed** and **MVVM** patterns.
- Built multi-currency checkout (PayPal, Stripe, Visa, MasterCard), phone-number authentication, proximity search and multilingual content flows for travel and local-business products.

**Programmer Club Leader — MDQ Academy, USA** Dec 2024 – Aug 2025

*Remote, part-time*

- Taught weekly programming fundamentals to students aged 10–14; designed lesson plans and coached debugging and problem-solving strategies live in class.

**Junior Flutter Developer — Oday Development Foundation** Mar 2024 – Jun 2024

- Implemented designs, wrote tests and resolved defects alongside senior developers on shipped Flutter applications.

**Earlier:** Real Estate Broker, In-Home (Jul 2020 – Feb 2022) · Marketing Intern, Careem & Uber (Apr 2018 – Nov 2019)

## PROJECTS

**Egypt Consulate Apps — Kuwait, Riyadh, Jeddah & Milan**

*Flutter · Flavors · Deep Links · Firebase · Google Play Console · App Store Connect | 8 live store listings*

*Kuwait: Google Play · App Store | Riyadh: Google Play · App Store*

*Jeddah: Google Play · App Store | Milan: Google Play · App Store*

- Developed and **published four separate consulate apps from one Flutter codebase** using flavor-based builds — three Gulf missions and one in Europe — each with region-specific feature sets, branding and store listings. Together they have 310K+ Google Play downloads; Riyadh is rated 4.5 stars from 2.2K+ reviews.

**New Courier App**

*Flutter · Google Maps · SQLite · FCM · Platform Channels | Private distribution, Android*

- Built a delivery-tracking app with **Google Maps**, real-time location tracking and **offline support via SQLite**, syncing against REST APIs when connectivity returns.
- Implemented **FCM with deep-linked actionable notifications** that route correctly even from a cold start, plus native location and camera modules over platform channels.

**Alkharj Business Guide — Mobile App & Web Admin Panel**

*Flutter · BLoC · Freezed · MVVM · Firebase | Live on both stores*

*Google Play · App Store*

- Local business and store directory for Alkharj city with category filters, **proximity search**, social links and Firebase phone-number authentication, backed by a responsive web admin panel for listing management.

**Remix — Short-Form Video Editor (Duet, Trim, Sound Reuse & Import)**

*Flutter · Cubit/BLoC · FFmpegKit · CameraX · video_player · image_picker | Private repository — walkthrough on request*

- Built a TikTok-style remix toolkit — **duet, trim and sound reuse** — as Cubit state machines with **FFmpeg** compositing; removed a **duet audio echo** and made scrubbing **lag-free** (throttled seeks, 1 s keyframes).
- Added **0.5×–3× speed** baked into exports (setpts/atempo), **gallery import** with a 4-minute limit, saved display modes and a filterable library with **share_plus** sharing; refactored to **sealed Cubit states**.

**RehlatiUAE**

*Flutter · BLoC · Freezed · MVVM · Stripe / PayPal | Shipped to Google Play; listing since retired*

- Event booking for UAE & Singapore travellers — PayPal, Stripe and card payments across currencies and languages.

## EDUCATION & ADDITIONAL

**Education:** Bachelor of Tourism & Hotels — High Institute EGOTH, Alexandria (Grade: Very Good)

**Languages:** Arabic (Native), English (Professional working proficiency)

**Certifications:** Flutter Advanced: Bloc & MVVM · Complete Flutter & Dart Course (2022) · Complete Dart Learning Guide

**Volunteering:** Student Union President, Faculty of Tourism & Hospitality (2018–2019) · Resala Charitable Society (2009–2017)